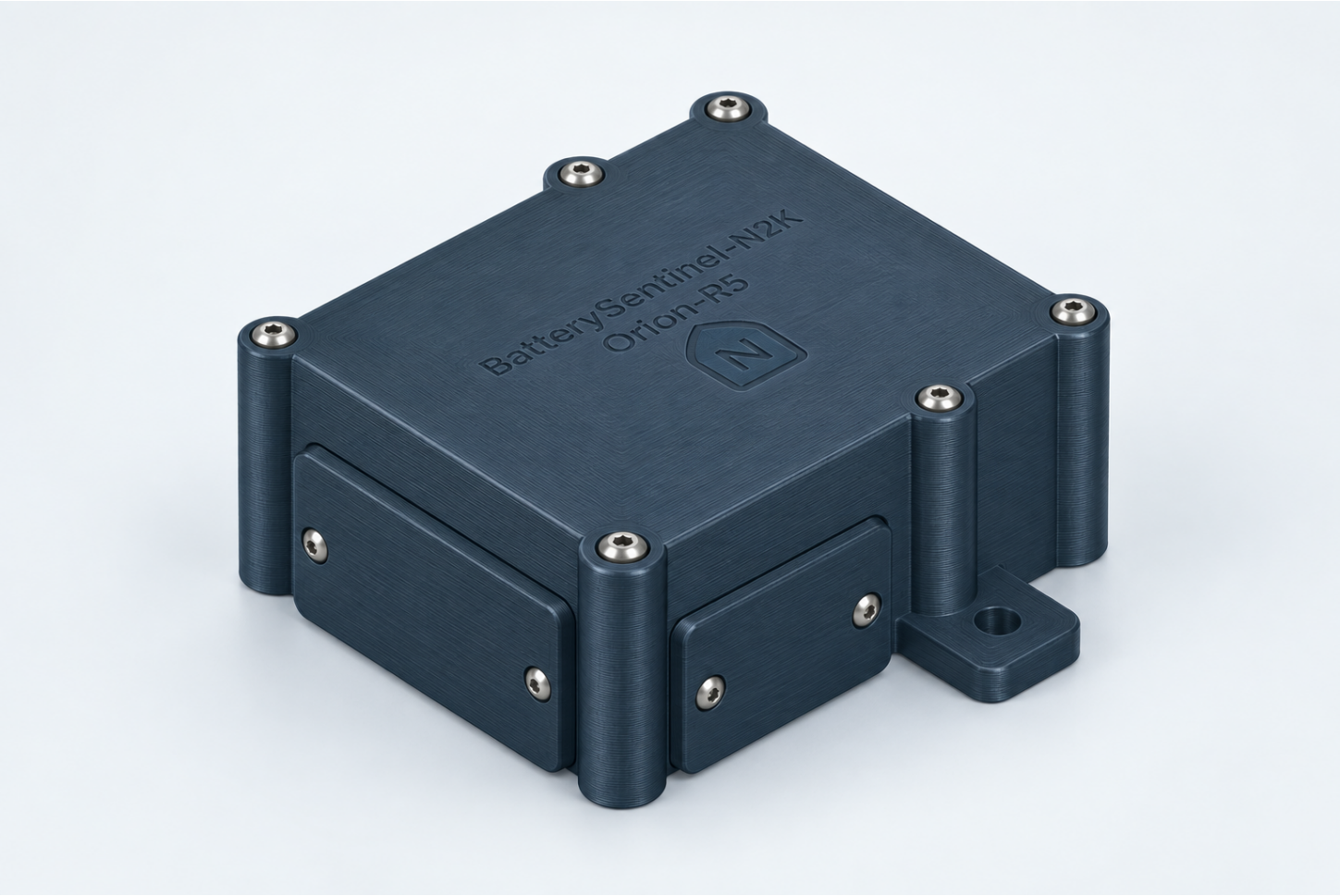
Enclosure
assembly guide

Build the enclosure. Understand its seals. Check the finished assembly.



For the prepared Bambu P2S 0.4 mm project

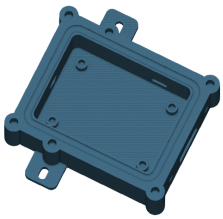
PETG shell and screw-on covers. TPU face gaskets and replaceable cable cartridge. Independent PCB mounting.

Start with the small fit coupon

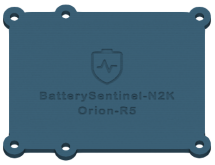
This is a printed prototype. Complete the empty-case leak and cable-retention checks before installing electronics.

Identify the printed parts

REFERENCE / KEEP THIS PAGE BESIDE YOUR WORKBENCH



A Base



B Lid



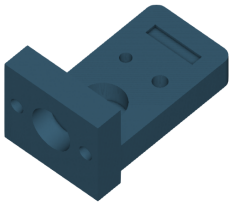
C USB cover



D Switch cover



E Pressure cap



F Fit coupon

PETG: A Base · B Lid · C USB cover · D Switch cover · E Cable pressure cap · F Fit coupon. Parts are individually scaled for identification.



G Lid gasket



H USB gasket



I Switch gasket



J Blank



J 5.0 mm



J 5.5 mm



J 6.0 mm



J 6.5 mm



J 7.0 mm

TPU: G Lid gasket · H USB gasket · I Switch gasket · J Cable cartridge: choose the blank or one of the five sleeve sizes.

One cartridge per enclosure

The six J alternatives are a solid blank and sleeves for nominal 5.0, 5.5, 6.0, 6.5 and 7.0 mm cable jackets. The coupon is a test piece; it is not installed.

Prepare hardware and prints

STEP 1 / WORK WITH THE ENCLOSURE DISCONNECTED

Hardware	Qty	Use
M3 x 8 mm screws	12	6 lid + 2 USB cover + 2 switch cover + 2 cable cap
ruthex RX-M3x5.7 inserts	12	Blind M3 pilots in the base
M2 x 5 mm screws	4	PCB only; screw-head diameter at most 4.0 mm
ruthex RX-M2x4 inserts	4	Four PCB standoffs
Suitable mounting screws + washers	2 sets	External ears; 3.6 mm wide slots

Use corrosion-resistant, non-countersunk screws, measured under the head. The list excludes spare inserts consumed by the coupon. Prepare calipers, an insert-setting tip and soldering iron, screwdrivers, deburring tools, and absorbent paper.

Disconnect NMEA power and USB before assembly. Selecting SERVICE alone does not remove all power from the device.

1

Print a trial before the whole enclosure

Open **Orion-R5-P2S-flush-colour.3mf**. Plate 3 contains **F fit_coupon** and **E cable_pressure_cap** in PETG; plate 4 contains a 6.0 mm J trial sleeve in TPU. Measure the actual cable and select the matching sleeve before assembly.

2

Keep materials and orientations separate

Use saved 0.20 mm layers and a 0.4 mm nozzle. Keep the two PETG lid parts together and map both colours to the actual spools. PETG uses 5 walls and 35% infill; TPU uses 100% infill and the external spool. Check filament temperatures and re-slice.

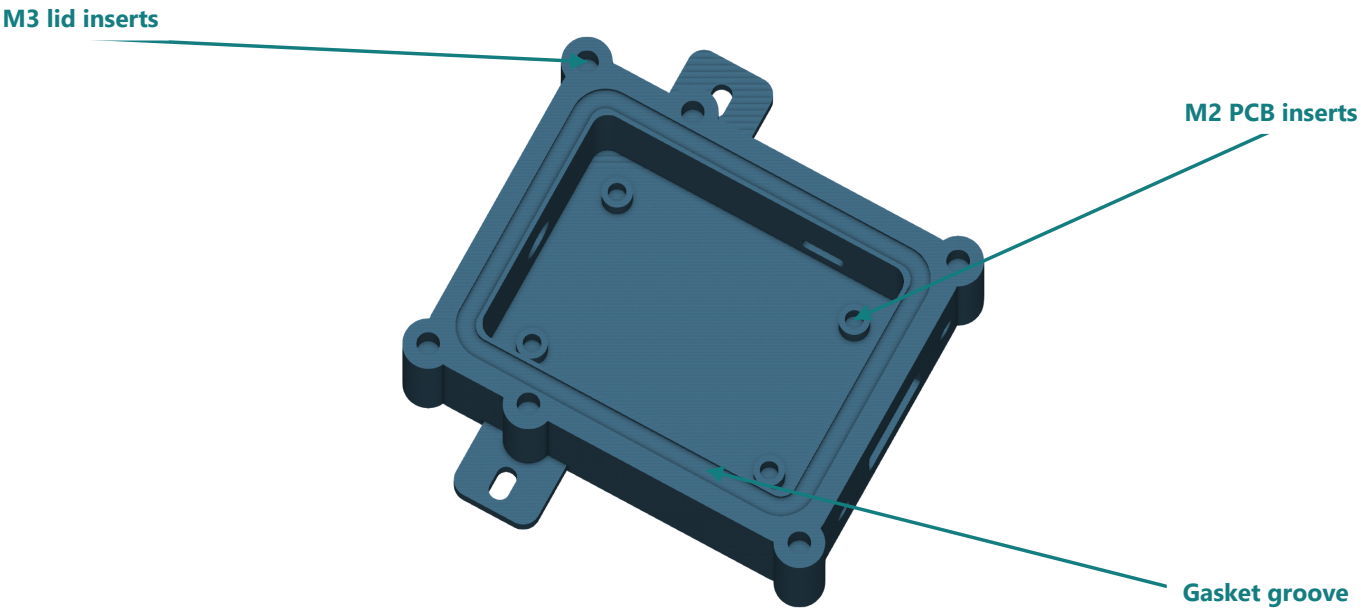
3

Inspect the parts

Sealing faces must be flat and clean. Reject gaps, cracks, warped covers or incomplete TPU loops. Check the roofs of the sideways holes for sagging. Keep support scars and raised seams off sealing lands.

Install the threaded inserts

STEP 2 / NO PCB, NO POWER, NO CABLE ATTACHED



Six lid insert positions surround the seal loop. Four smaller standoffs inside support the PCB. The six sideways M3 pilots serve the two covers and cable cap.

1 Prove the insert fit on F

Test both insert sizes and the horizontal pilots. Choose the tool temperature for the insert and filament using the manufacturer guidance and the coupon. No universal temperature or tightening torque is specified.

2 Set every insert straight and flush

Support the part, heat the insert and guide it squarely into its pilot. Let the softened plastic accept the insert without forcing it. Stop flush with the seating face, keep the tip clear of the blind bottom, and let it cool completely.

3 Check threads and closed bottoms

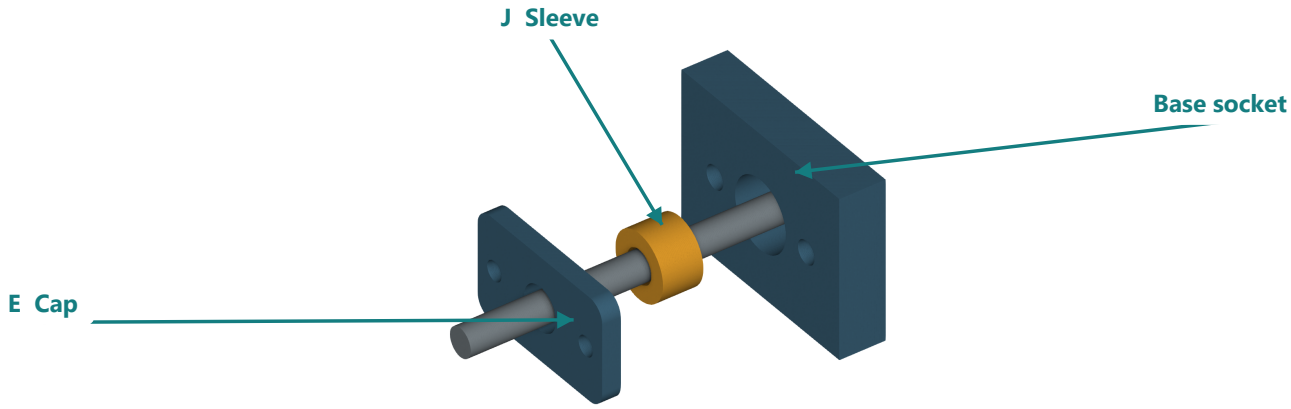
Start each correct screw gently by hand. An insert must not spin, tilt or stand proud. Remove debris from the grooves and cavity. Do not drill the pilots through.

Pilot sizes are starting dimensions

M3: diameter 4.0 x depth 6.7 mm. M2: diameter 3.2 x depth 4.6 mm. The real print and heat-setting process determine the fit.

Thread and seal the cable

STEP 3 / BEFORE ANY CABLE SOLDERING



Outside to inside: **E pressure cap** → **J sleeve** → **base entry**. The cropped wall stays intact in the printed base. Cable and expanded spacing are illustrative; no cable diameter is established by this image.

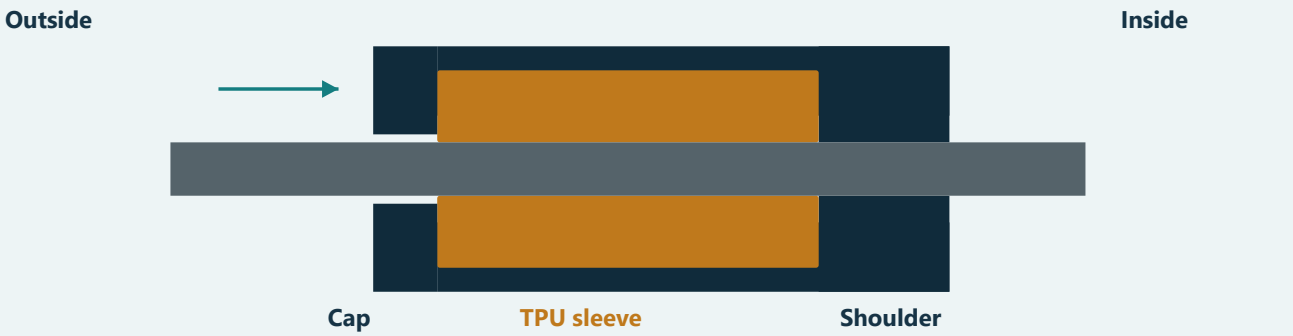
1 Measure the intact jacket

Measure at several positions and directions without flattening the cable. The HangTon cable has no verified jacket diameter yet. Pick a suitable sleeve after delivery; use the blank if no cable is installed.

2 Feed the bare end through all three parts

Pass the bare-wire end through E, then through the unsplit J sleeve, and into A. Keep the intact, round jacket through the full sleeve and into the cavity. The molded connector cannot pass through the small entry.

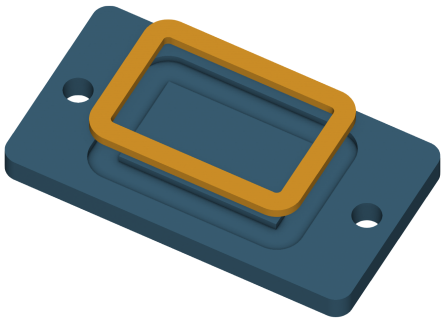
CABLE SEAL: ONE UNSPLIT CARTRIDGE

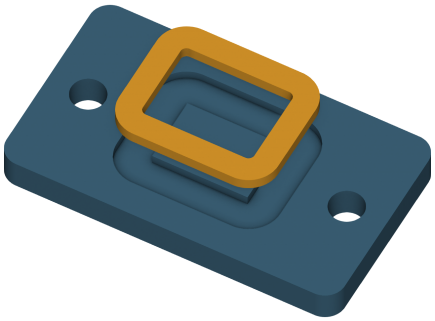


The cap compresses J axially from **7 to 6 mm** against the inner shoulder. This confines the sleeve around the jacket. Seat the cap with two M3 x 8 screws, alternating between them. Stop when the rigid faces meet. Grip and leak performance must be tested; extra tightening cannot correct the wrong sleeve.

Close and test the empty case

STEP 4 / COMPLETE WET TESTS BEFORE ELECTRONICS





C + H: USB cover and its gasket. **D + I:** switch cover and its gasket. Each gasket goes in its **cover groove**; each cover uses two M3 x 8 screws.

1

Fit the covers and lid without the PCB

Lay each TPU loop flat, with no twists or debris. Fit both service covers. Place G in the **base** lid groove, then fit B with six M3 x 8 screws. Bring the rigid lands together evenly; see page 8 for the pattern.

2

Check the intended splash exposure

Place absorbent paper in the empty case, clear of the seals. Close it with the actual cable and selected cartridge, or the blank. With all wiring disconnected, check the expected splash exposure. Dry the exterior before opening; inspect the paper, walls, seams and cable entry for moisture.

3

Check retention, then repeat the leak check

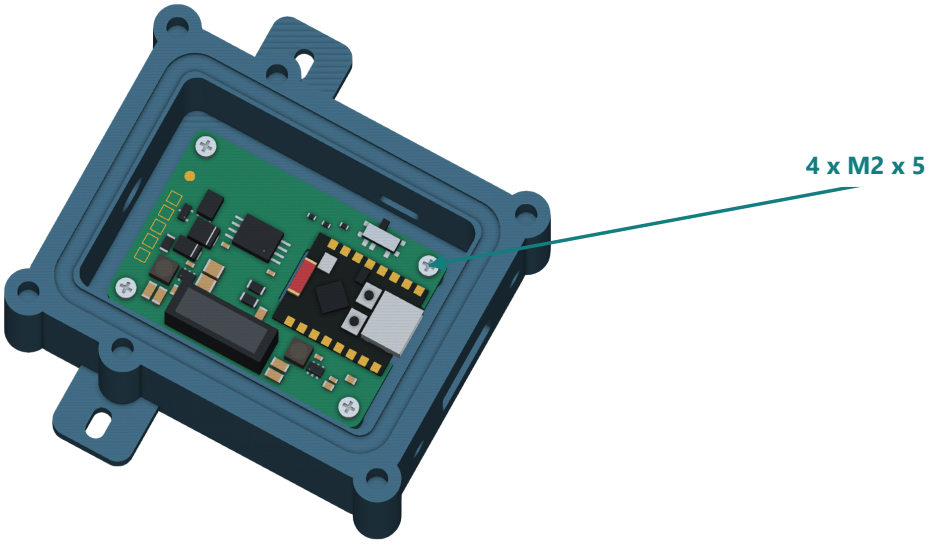
Apply representative cable pulling and flexing without damaging the jacket. The jacket must not creep through the sleeve. Repeat the empty-case leak check after flexing and after opening and reclosing the covers. Record the test conditions and result.

If moisture enters, stop here

Identify and correct the leak, then repeat the empty-case test. Dry everything completely before installing electronics. If a seal changes later, remove electronics before another wet test.

Install the electronics

STEP 5 / ONLY AFTER THE MECHANICAL TRIAL PASSES



The current 52 x 36 mm R5 board mounts on four internal standoffs. Four illustrative M2 screw heads show PCB retention; lid screws do not clamp the board.

1 Confirm the fit, then terminate the cable

Keep power disconnected. Dry-fit the board, actual USB plug and switch access before soldering. The cap and unsplit sleeve must already be on the cable. Connect conductors using the current R5 PCB labels and wiring documentation, with continuity checks.

2 Position the PCB and leave wire slack

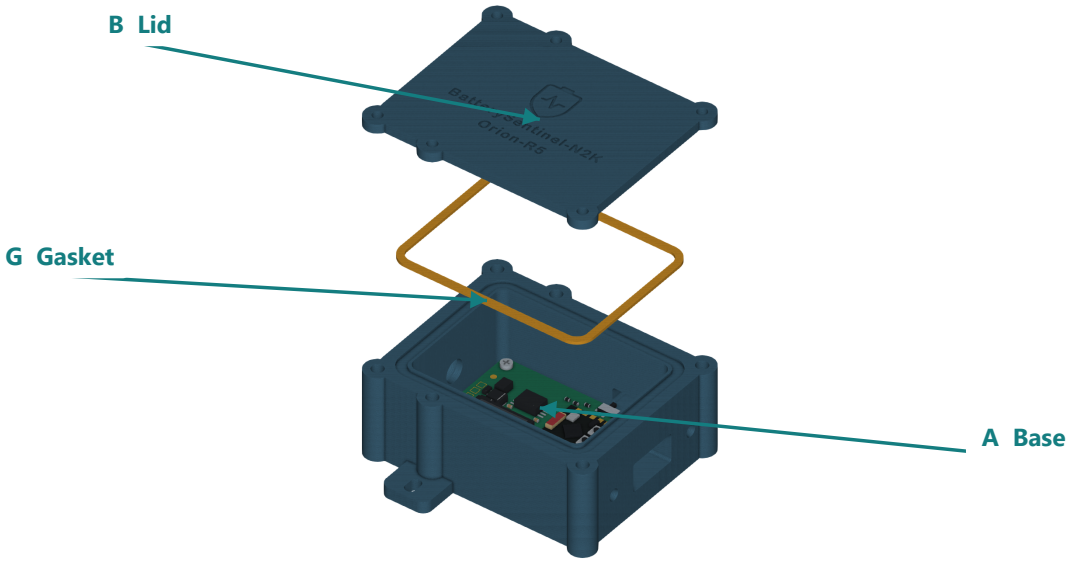
Confirm the correct USB and switch sides. Keep the intact jacket through the entire sleeve; strip only inside the cavity, clear of the seal. Keep solder tails clear of the floor and conductors clear of standoffs, the antenna region and sealing grooves. Leave enough wire slack to avoid pulling on solder joints.

3 Retain the PCB with four M2 x 5 screws

Start all four screws before tightening. Seat them gently without bending the PCB. Use M2 x 5 screws only at these four supports. With a 1.6 mm PCB and flush inserts, nominal engagement is 3.4 mm and blind-tip clearance is 1.2 mm.

Seat the lid gasket and close

STEP 6 / CONTROLLED COMPRESSION, NO TRAPPED WIRES



G sits in the groove in A. The short lip inside B locates the lid. All six lid screws are outside the continuous gasket loop.

HOW THE FACE SEAL WORKS

Before tightening



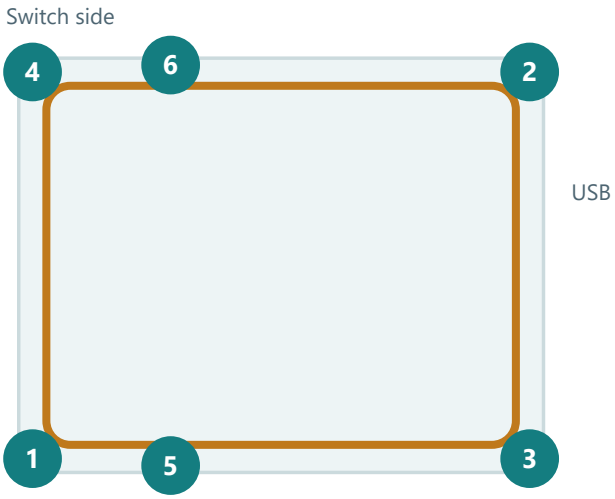
Free height: 1.2 mm

Rigid lands meet



Closed gap: 1.0 mm

The free face gaskets are **1.2 mm thick**. The closed gap is **1.0 mm**: about **16.7% compression**. The groove leaves room for sideways expansion. This section is schematic; the real gasket must remain a complete loop.



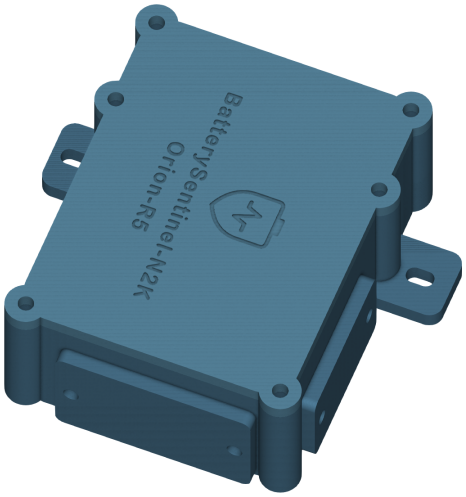
Tighten in the numbered order

Start all six M3 x 8 screws, then make several gentle passes. Alternate across the lid. Stop when the rigid lands meet evenly; do not bow the lid or force an insert.

Refit both service covers and confirm that every seal is seated.

Mount and use the service covers

STEP 7 / KEEP ACCESS DRY



Assembled CAD view with blank cartridge; fasteners omitted. Nominal envelope: 82.4 x 88 x 33.2 mm, excluding screw heads and cable. Mounting-slot centres are 76 mm apart, outside the sealed cavity.

1 Fix through the external ears

Choose a sheltered location with service access and adequate radio reception. Use fixings and washers suited to the mounting surface. Support both ears without twisting the case; do not drill through the enclosure floor.

2 Support and route the cable

Anchor the cable near the case so its weight and motion do not load the printed seal or solder joints. Avoid sharp bends and chafing. Route it so water runs away from the entry. Confirm the cable and case stay stationary during normal movement.

USB service sequence

In a dry location, remove the switch and USB covers. Select **SERVICE before connecting USB**. After service, **disconnect USB before selecting RUN**. Do not assume a left/right switch direction. Refit both gaskets and covers before boat operation.

Check the real USB plug: its body must fit through the opening and reach the recessed socket fully. Never force a plug against the housing. Verify temperature and Wi-Fi/Bluetooth reception with the final closed case at its installation location.

Inspect, troubleshoot and maintain

CARE / RECHECK AFTER SERVICE OR SEAL CHANGES

Symptom	Likely check	Action
A cover will not sit flush	Twisted gasket, print debris, proud insert or trapped wire	Remove the cover. Correct the cause; never force the screws to pull it flat.
A screw stops early or spins	Wrong length, poor thread start, tilted or loose insert	Remove the screw. Confirm size and insert seating. Repair or reprint the part and repeat fit checks.
Cable slides or jacket is damaged	Wrong sleeve size, uneven cap seating or excessive external load	Measure again, select/rebuild the sleeve and support the cable. Repeat retention and empty-case leak tests.
Water appears inside	Seam, printed wall, damaged gasket or cable entry leak	Disconnect and remove electronics. Find the entry path and correct it. Repeat testing with an empty case.
USB does not seat fully	Plug overmould or reach does not match the opening	Check the actual plug and socket. Use a compatible plug or revise the opening; do not force it.
Poor radio reception or heat buildup	Installation location or closed-case operating conditions	Review location and cable routing. Validate reception and temperature before permanent use.

After opening, inspect each gasket for cuts, permanent flattening, stretching and dirt. Replace a damaged loop as a complete part. Keep the mating faces clean and flat. Check that all covers seat evenly and inserts remain secure.

No waterproof or IP rating has been established

The CAD and slicer checks establish printable geometry. FDM wall porosity, actual TPU behaviour, salt exposure, vibration, heat and ageing still need physical evaluation. Use a qualified enclosure or revise the sealing system if this prototype cannot pass the required exposure.

Assembly and acceptance record

KEEP THIS PAGE WITH THE DEVICE

- [ ] Coupon checked with the actual PETG, TPU, inserts and screws.
- [ ] Cable jacket measured; one sleeve or blank selected and recorded.
- [ ] All 16 inserts straight and flush; no blind pilot drilled through.
- [ ] Empty-case leak test passed with the intended cable or blank.
- [ ] Cable-retention/flex test passed; leak test repeated afterwards.
- [ ] Case fully dry; PCB fitted with four M2 x 5 screws and free wire slack.
- [ ] Three face-gasket loops seated; six lid screws and six cover/cap screws fitted.
- [ ] Closed-case temperature and radio reception checked at the installation site.

Cable OD / cartridge: _____

PETG / TPU brand and grade: _____

Test exposure / duration: _____

Test date / result / assembler: _____

Files for this edition

Orion-R5-P2S-flush-colour.3mf contains four plates: rigid parts, TPU alternatives, PETG trial pieces and a TPU trial sleeve. The lid is one object with two colour parts. Install five rigid pieces, three face gaskets and one cartridge; F remains a test piece.

Technical references

Use the enclosure README and sealed-contract.json beside the 3MF for dimensions, part selection and hardware. For electrical termination and operation, use the project's Orion-R5 operating and installation manual.

ruthex RX-M3x5.7 insert guidance · RX-M2x4 guidance
Prusa: watertight printed enclosures and seal limitations

Illustrations: generated cover based on the enclosure reference; technical views rendered from the final CAD model. Metal fasteners are illustrative. Seal sections are enlarged schematics. Dimensions and instructions take priority over apparent image scale.